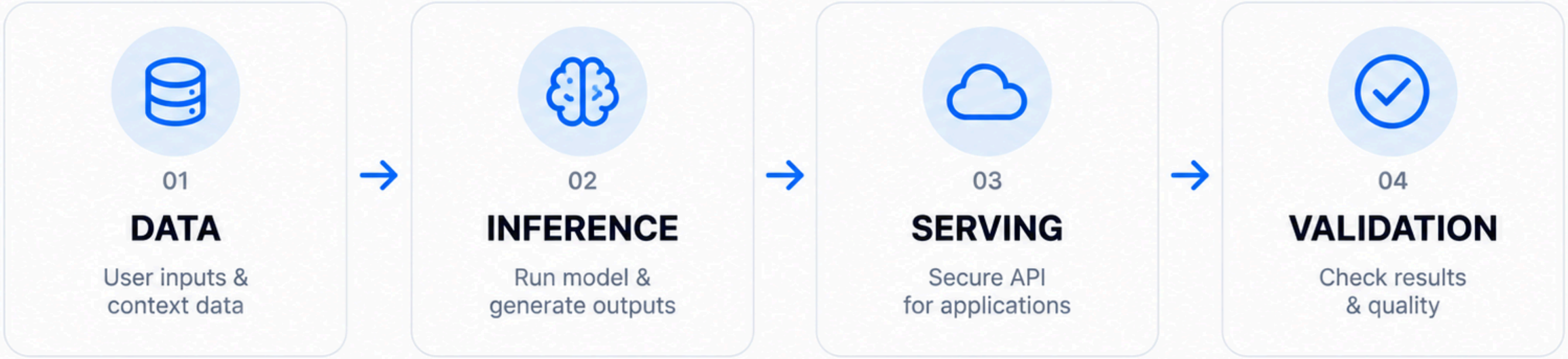
GeoLife — Home/Office Inference & MLE Serving

Track B1 · Final Demo · 20.10.2026



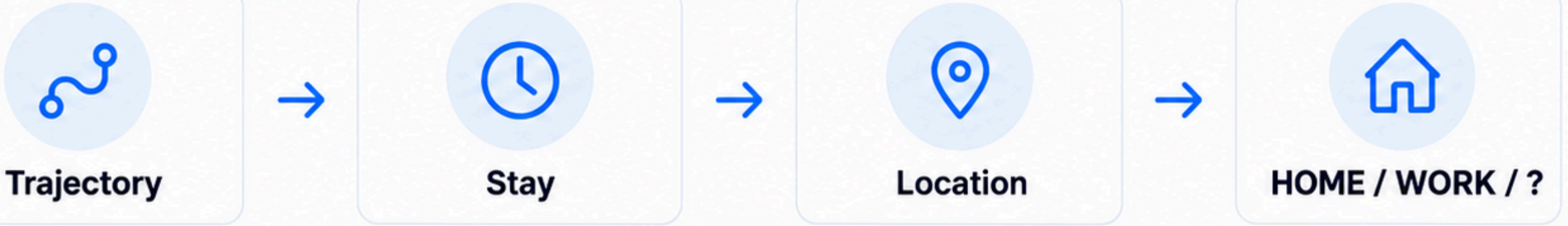
DATA → INFERENCE → SERVING → VALIDATION

Problem & Scope: No True Home/Office Labels

Input
GPS trajectory .plt history, UTC timestamps, uneven sampling.

Objective
detect stays → recurring locations → HOME / OFFICE / abstain.

Constraint
no semantic ground truth; heuristic confidence ≠ accuracy.



Explainable rules + abstention when evidence is weak.

Project Journey: 16/09 → 20/10

Research and MLE progressed in parallel.

16–18/09 · CP1

- ingest & EDA
- cleaning
- stay detection + OpenAPI

18/09–15/10 · CP2 & Research

- timezone-v2 + complete-link
- HOME / OFFICE inference
- robustness, comparators, Evidence-v2 audit

12–20/10 · CP2/3 MLE

- Docker / ECR / EC2
- canary–rollback
- Cognito + CloudFront, SQS job E2E

20/10
current demo milestone

Two tracks ran together: research quality + deployment validation.

GeoLife Raw Dataset: 24,876,978 GPS Points

Large data volume, uneven user coverage, and no HOME/OFFICE ground truth.

182
users

18,670
.plt files

24,876,978
GPS points

Data distribution is highly skewed across users.

Only 69 user folders contain transportation labels — there are no HOME/OFFICE labels.

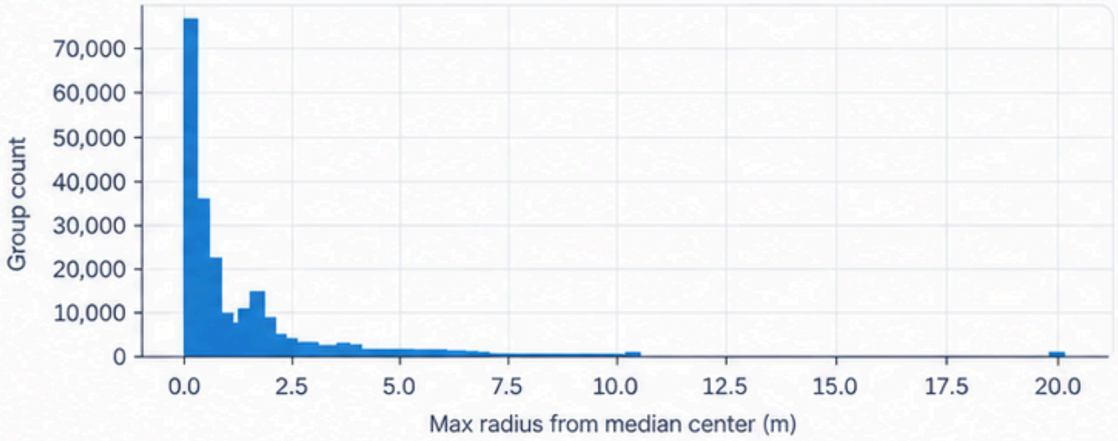
Top-10 users contribute about 47.4% of trajectories.

EDA Changed the Preprocessing Rules

What we learned from duplicate timestamps and temporal gaps.

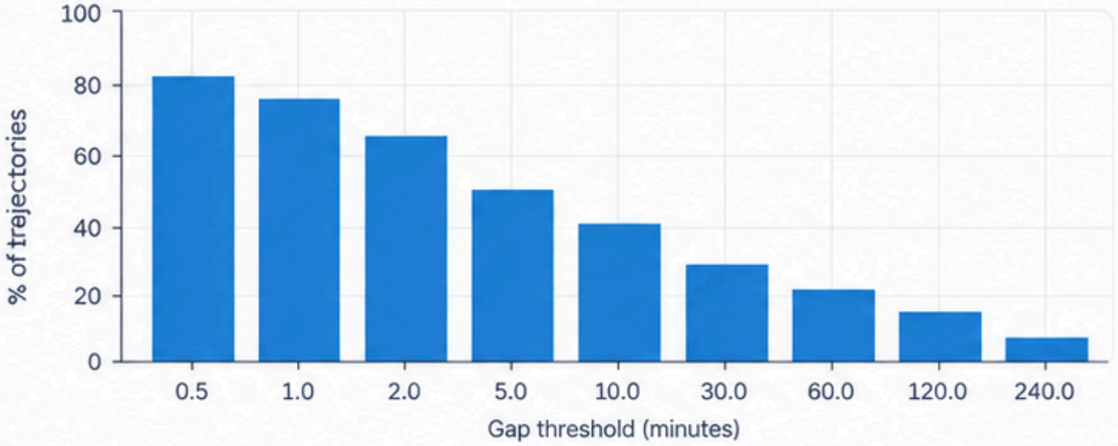
Same-second spatial jitter

Distribution of max radius within each same-second group (clipped at 20 m)



Temporal gaps between points

Share of trajectories with at least one gap larger than threshold



698,900
duplicate-timestamp rows (2.81%)

99.61%
of same-second groups stay within 10 m

50.95%
of trajectories have a gap > 5 minutes

1,677
files belong to exact-duplicate groups

Do not connect every GPS point blindly — continuity must be broken when evidence is weak.

CP1: Cleaning → Stay-Point Detection

A frozen engineering baseline before semantic HOME/OFFICE inference.

Key parameters (engineering baseline)

max_gap = 300 s
Split sequence if time gap > 300 seconds

distance = 200 m
Points within 200 meters are considered close

min_dwell = 20 min
Minimum time to qualify as a stay

- Sensitivity grid: 27 configurations (3 × 3 × 3).
- Thresholds are engineering tolerances, not proof of semantic optimality.

Stay detection runs only within continuous sequences.

CP1 Results: Coverage Waterfall

From raw GPS observations to stay-bearing users under the frozen baseline.

24.88M
raw points

GPS observations after preprocessing (duplicate and invalid points removed).

5,821
stays

Stay segments detected under the frozen engineering baseline.

136
stay-bearing users

Users with at least one detected stay (candidate for HOME/OFFICE inference).

46 / 182
users have no stay under the current baseline.

Abstention begins at the observational-data stage, before semantic HOME/OFFICE inference.

These are **offline results on the real GeoLife dataset**, not Cognito demo users.

Parity contract: keep 5,821 stays stable instead of relaxing thresholds to force full coverage.

CP2: Timezone-v2 + Recurring-Location Clustering

Local-time behavioral windows and compact recurring-location clusters.

5,821 / 5,821
stays resolved to IANA timezone

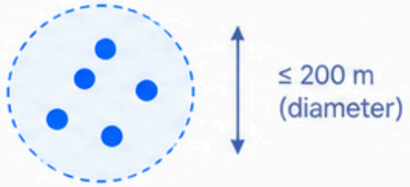
Use local time per stay, not a fixed UTC+8 assumption.

2,015 locations
716 recurring
104 users with recurrring locations

Complete-link 200 m
guarantees cluster diameter ≤ 200 m

Complete-link 200 m

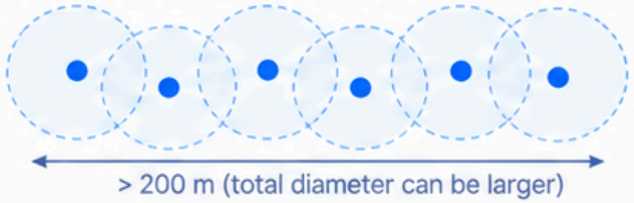
Compact, tight clusters with a hard radius bound.



hard geometric bound

DBSCAN eps=200 m

Can form chains through nearby points.



can chain neighbors beyond 200 m total diameter

Complete-link chosen because 200 m is a true geometric constraint, not just local neighbor reachability.

Timezone lookup is performed for every stay before applying HOME/OFFICE local-time windows.

CP2: HOME/OFFICE is Evidence, Not Probability.

Rule-based emission with abstention when behavioral support is weak.

27
HOME

16
OFFICE

37
unique emitted users

HOME window: 21:00–06:00 local
Use interval overlap, not only arrival hour.

OFFICE window: weekdays **09:00–17:00** local
Use interval overlap, not only arrival hour.

Heuristic evidence score

Dwell share
Fraction of time in the candidate location.

Top-two margin
Difference between top-1 and top-2 location shares.

Support dates
Number of distinct days supporting the location.

HEURISTIC EVIDENCE SCORE

evidence_strength = (share + margin + min(days/5, 1)) / 3

If evidence is insufficient, the system abstains instead of forcing a label.

A score like 0.8 is not an 80% probability of correctness.

HOME and OFFICE use different gates; OFFICE is weaker and more schedule-sensitive.

Robustness: DBSCAN Chaining & Independent Comparators

Hard geometric constraints matter, and comparator agreement is not ground truth.

Historical 97-user cohort

DBSCAN 200 m

68

clusters diameter > 200 m

max diameter **836.66 m**

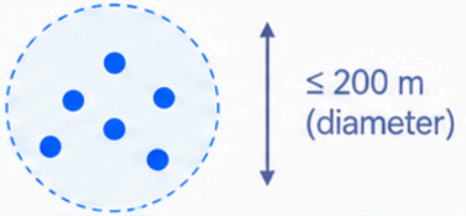
> 200 m (total diameter can be much larger)

Complete-link 200 m

0

clusters diameter > 200 m

max diameter **199.23 m**



compact cluster within 200 m bound

Independent comparator: Trackintel OSNA semantic-only

HOME: 27 / 27 agreement

OFFICE: 11 / 16 agreement

Comparator agreement with the frozen result is NOT classification ground truth.

DBSCAN can chain neighbors beyond the intended spatial bound; complete-link preserves the 200 m geometric constraint.

Research Beyond Baseline: Expand, But Don’t Overclaim

Research tiers explored broader coverage, while production stayed frozen and evidence-v2 was not promoted.

Reliability
Stage 05
Data quality, stability, and robustness checks.

Routine / work profiles
Stage 06
Different user segments and work routine patterns.

POI historical context
Stage 07
Temporal context and historical place information.

Trackintel & literature comparison
Stage 08
Comparison with Trackintel and related literature.

Research tiers
Expanded offline coverage

36 HOME

19 OFFICE

Research results are not automatically promoted.

Frozen production baseline
Current baseline kept for production

27 HOME

16 OFFICE

Evidence-v2 audit result

21 / 55 candidates
enough inputs to compute v2.

Useful holdout-matched comparisons were very limited.

Do not promote Evidence-v2; freeze the baseline.

Research can increase insight and exploratory coverage, but production claims must stay tied to verified evidence.

API Contract & Model Variants

Clear interfaces, explicit abstention, and transparent separation between private GeoLife inference and public synthetic demo paths.

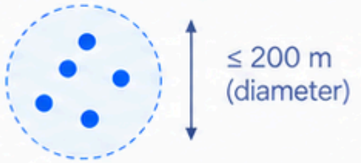
OpenAPI / API contract

POST	/classify
GET	/health
GET	/version

- typed request / response schema
- versioned interface
- abstain is a model output, not an HTTP failure

Model variants under validation

Baseline — complete-link 200 m



hard geometric bound

Candidate — DBSCAN 100 m

density-based, more flexible shape

Used for validation and comparison, **not silently mixed.**

Important boundary

Private GeoLife model path

Inference on real GeoLife data, producing semantic labels.

stays / recurring locations → HOME | OFFICE | abstain

≠ These are NOT the same inference result.

Public synthetic Top-3 demo

Uses synthetic data for demonstration, not real user inference.

authenticated app → destination suggestions (Top-3 locations)

A stable API contract is not just about endpoints — it must preserve model semantics and deployment boundaries.

Checkpoint 2: Docker → ECR → EC2 + Canary / Rollback

Internal AWS sync demo validated through SSM-only access and synthetic HTTP checks.

Docker image
build and tag
digest-pinned container

→

ECR
push image
by digest (immutable)

→

EC2 FastAPI
pull and run
container on EC2

→

SSM access
manage and validate
via AWS SSM (no SSH)

digest-pinned container, no public EC2 inbound

Baseline
50 / 50
successful
complete-link 200 m

→

100% DBSCAN Canary
50 / 50
successful
candidate variant

→

Rollback to Baseline
10 / 10
successful
recovered baseline routing

p95 latency (ms)
client-observed

Baseline	Canary (100% DBSCAN)	Rollback
157	273	201

Scope
Synthetic requests through an internal SSM-only experiment.

Boundary
100% canary on one synthetic account does not prove 90/10 production behavior.

Verified: Baseline → Canary → Rollback all passed on AWS.

Current Deployment: Private Network, Public HTTPS

Public web access at the edge, private model-serving infrastructure inside AWS.

Web app & inference (synchronous)

Public HTTPS access, private network for model serving

React SPA / CloudFront
private S3 (static assets)

→ **Cognito**
Hosted UI + PKCE

→ **JWT API Gateway**
authorizer

→ **VPC Link**
private integration

→ **Internal ALB**
private subnet

→ **EC2 / FastAPI**
model inference (private subnet)

Async jobs (batch processing)

Synthetic data generation and long-running jobs

POST /v2/jobs
authenticated (returns job ID)

⇢ **Private S3**
synthetic payloads

DynamoDB
job state

SQS
queue + DLQ

⇢ **Bounded EC2 worker**
process jobs (private subnet)

⇢ **Polling / job status**
GET /v2/jobs/{id} (owner-scoped)

SQS carries pointers / job IDs, not raw GPS coordinates in the message body.

Security

EC2 has no public inbound; model API is protected and jobs are owner-scoped.

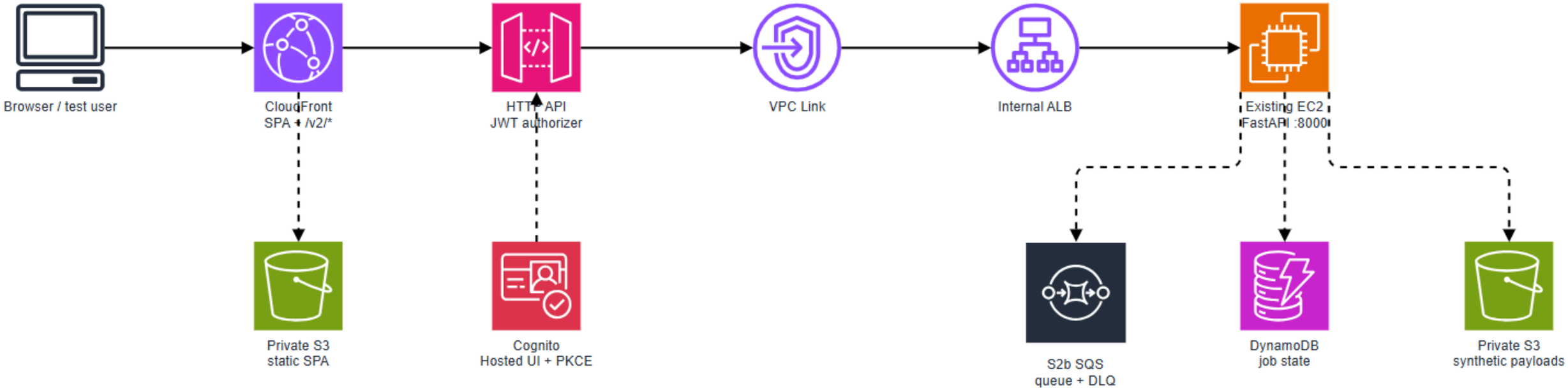
Operations

ECR immutable tag / manifest digest retained for rollback.

Public HTTPS for the app; private network boundaries for serving and jobs.

GeoLife Mentor Beta — demo-day architecture

Read-only checkpoint | 2026-10-20 | Existing deployment



Solid = live HTTPS request path. Dashed = auth / synthetic async storage path.

One synthetic S2b E2E PASS; worker exited 0, recurring timer DISABLED. Concurrency and retry/DLQ NOT TESTED.

Live Demo: Cognito Login → Top-3

Authenticated public demo flow using synthetic recommendation data.

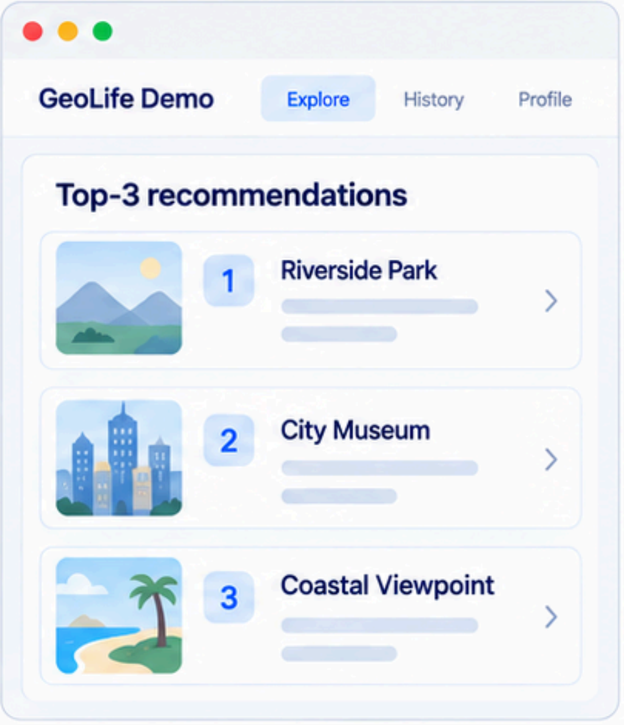
https://dxyhw0cktlmvt.cloudfront.net/

01
Open CloudFront SPA
Access the public demo web application.

02
Login with Cognito (code + PKCE)
Authenticate via Amazon Cognito.

03
JWT-protected API call
Call secured endpoints with access token.

04
Top-3 destination suggestions
Receive synthetic Top-3 results.



Manual owner verification
Login + authenticated Top-3 **PASS.**

Important boundary
Recommendations are synthetic-only, not real GeoLife GPS inference.

If live login is slow
Show protected-route behavior and move to saved E2E evidence.

Demo proves authentication + serving path, not HOME/OFFICE accuracy.

Checkpoint 3: S2b ONE Controlled AWS Async Job PASS

One synthetic AWS async job completed end-to-end with owner-scoped access control.

Cognito User A
authenticated
(owner-scoped)

→

POST 202
submit async job
(returns job ID)

→

SQS (1 → 0)
message enqueued
and consumed

→

Worker
processes job
(private subnet)

→

DynamoDB
succeeded
(job status updated)

→

GET poll
retrieve result
(job succeeded)

Queue wait
141.3 s
includes intentional waiting while worker was off

Processing
3.53 s

Server-side E2E
144.8 s
1 job only; not p95

User B GET User A's job → 404
owner-scoped isolation worked

Worker exit 0
worker completed successfully

ALB healthy
target group healthy during run

Verified: POST 202 → queue → worker → succeeded → poll result, with tenant isolation.

Research Integrity & Security: What Is Verified vs Not Claimed

Separate what has been demonstrated from what remains outside today's evidence.

VERIFIED

stay / cluster parity
Top-3 recommendations match between demo and synthetic dataset.

synthetic sync demo & one async job
End-to-end flow works with synthetic data (sync) and one PASS async job.

JWT 401 protection
API endpoints require valid JWT; unauthorized returns 401.

owner-bound User B 404
User B cannot access User A's data; returns 404.

EC2 private + ALB healthy
Model and worker run in private subnets; ALB health checks pass.

NOT YET VERIFIED

HOME / OFFICE accuracy
No real-user validation of HOME/OFFICE detection accuracy.

real GeoLife public inference
No inference using the actual public GeoLife dataset yet.

production latency / SLO
No measured performance, latency, or SLO targets.

concurrent load
No multi-user or sustained load testing.

retry / DLQ behavior
No validation of retry logic or DLQ handling under failures.

recurring worker
No long-running or scheduled/recurring worker execution yet.

full monitoring / drift dashboard
No CloudWatch/metrics dashboard or model drift monitoring.

One PASS async job is functional evidence, not a performance benchmark.

No raw GPS, private HOME/OFFICE locations, or tokens shown on slides.

Roadmap & One-Sentence Take-away

What comes next after the current offline research and AWS serving validation.

01

Real GeoLife artifacts

private versioned inference artifacts → v1/v2 serving

privacy / license gates

02

Operational MLE

worker scheduling + retry / DLQ + multi-user load + latency histogram + CloudWatch dashboard

03

Model evolution

shadow / canary 90:10 + chronological drift replay (simulated) + POI enrichment / independent evaluation

Next phase: connect private historical artifacts to serving safely, then validate operations with statistically meaningful measurements.

Take-away: Không chỉ deploy được API — hệ thống có data lineage, baseline kiểm chứng, và ranh giới evidence rõ ràng.